Training partition only

UK-DALE 1; REFIT 3, 5, 9, 11 · 50% recorded unchanged · 50% recomposed shown · validation/test recorded only

Step 1 — Focal source-window sampling

- Select focal appliance i^* uniformly
- Sample a source window with $\exists t: \; z_{t,i^\star} = 1$
- Other appliance traces sampled independently

$P_{t,i^\star}^{(k_{i^\star})}$ · focal trace with at least one ON sample

Step 2 — Median-residual bin sampling

B1	B2	B3	B4	B5
$[0, 100)$	$[100, 200)$	$[200, 400)$	$[400, 800)$	$[800, \infty)$ W

$\mathbf{u}_t = R_t^{(k_0)} + \sum_{i \neq i^\star} P_{t,i}^{(k_i)}$
independent other targets + stratified residual

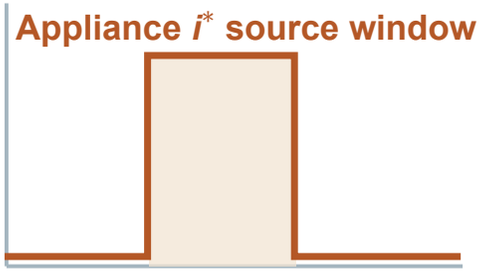
Recomposed training window: $\tilde{x}_t = R_t^{(k_0)} + \sum_{i=1}^{M} P_{t,i}^{(k_i)}$

For each appliance: $\tilde{P}_{t,i} = P_{t,i}^{(k_i)}$, $\tilde{z}_{t,i} = z_{t,i}^{(k_i)}$

Construction of one recomposed training window

(a) Focal appliance trace

Appliance i^* source window

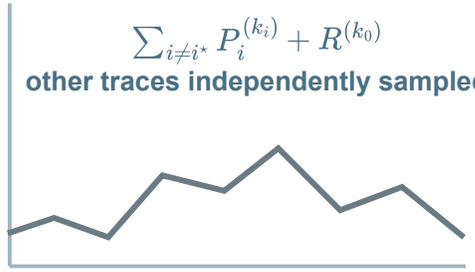


$\exists t: \; z_{t,i^\star} = 1$

+

(b) Other targets + residual

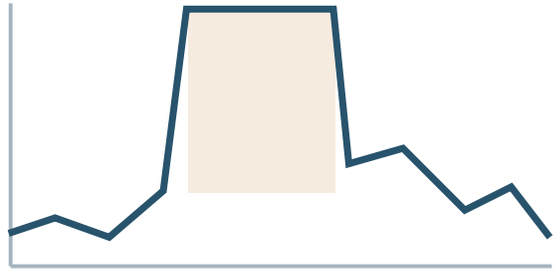
$\sum_{i \neq i^\star} P_i^{(k_i)} + R^{(k_0)}$
other traces independently sampled



Recorded residual selected by bin

=

(c) Composite Input $\tilde{\mathbf{x}}$



Selected targets and labels retained